

ZHONGHAO HE

zh378@cam.ac.uk | hezhonghao.github.io | github.com/hezhonghao | [Google Scholar \(1000+citations\)](#)

SUMMARY

I am Zhonghao. I work on AI alignment and human-AI interaction research. My previous work got accepted by NeurIPS, ICML, ACM FAccT, and ICLR (workshop), etc. My major interests are to build machines that help humans learn and think. Methodology-wise, I do formal modeling, algorithms, (post-) training, evaluation, simulation, and computational social science methods. Currently I focus on two things, to develop Martingale training as a principled method to address Bayesian irrationality such as confirmation bias, sycophancy, and polarization; and to develop alignment algorithm that seeks human reflective equilibrium. [Read my research proposal here.](#)

RESEARCH EXPERIENCE

Cosmos Fellow / Research Associate - Oxford *Mar 2026 - Sep 2026*

Admitted as Cosmos Fellow of HAI Lab, Oxford. Working with Prof Philipp Koralus of Philosophy Department and Prof Jakob Foerster of Engineering Department. Projects: Martingale training for Bayesian rationality, Coherence optimization for self-improvement, Coherence game for learning human reflective equilibrium, and infrastructure for truth-seeking AI. [Coherence Theoretical](#)

Research Engineer - CMU (Remote) *Jan 2025 - Dec 2026*

Co-lead “Martingale Score”: We introduce a Bayesian statistical method to evaluate confirmation bias in LLM reasoning, with Profs Maarten Sap & Hirokazu Shirado [Link to Martingale Eval Paper](#)

Research Engineer - University of Washington (Remote) *Oct 2024 - Jun 2025*

Co-led two papers: “The Lock-in Hypothesis”, and “Open Problems in AI Influence”, with Prof Max Kleiman-Weiner [The Lock-in Hypothesis Website](#)

Researcher - University of Cambridge *Dec 2023 - Jul 2025*

Worked on multiple projects on interpretability, alignment, and agentic safety, with Profs David Krueger, Yaodong Yang, Grace W. Lindsay, and Anya Ivanova.

EDUCATION

Johns Hopkins University (incoming) *Sep 2026 -*

PhD in Computer Science

Supervisor: Gillian K. Hadfield

Direction: AI Alignment, principled RLHF alternative

University of Cambridge *Sep 2022 - Jul 2025*

Mst in AI Ethics and Society

Supervisor: Henry Shevlin

Coursework: ML Safety, AI Alignment, AI Ethics, RL, Advanced DL, Algorithm and Data Structure, Mechanistic Interpretability, etc.

Stanford University *May 2019 - Aug 2019*

Cognitive Science Summer Semester

Courses: Mathematics Foundation of Computing, Minds and Machines, Introduction to Neuroscience

Shantou University *Aug 2014 - Jun 2019*

BA in English and Linguistics

Relevant Coursework: Linguistics, ML, Maths.

AWARDS AND GRANTS

Cosmos Fellowship <i>Yearly \$75,000, pro rata, plus \$25,000 team compute</i>	2026
UK AISI Alignment Project Grant <i>£290,000; Team</i>	2026
Cosmos Grant on Truth-seeking AI <i>\$120,00 Team compute budget</i>	2025
Foresight Institute AI Safety Research Grant <i>\$40,000; Team</i>	2025
Lambda Research Grant <i>\$5,000; Team</i>	2024
Manifund Research Scholarship <i>\$5,000; Individual</i>	2023
Open Philanthropy's Graduate Scholarship <i>£50,000 to support master study</i>	2022

PUBLICATIONS

- [1] **Z. He***, T. Qiu*, H. Shirado, M. Sap (2025) Stay True to the Evidence: Measuring Belief Entrenchment in LLM Reasoning via the Martingale Score. *NeurIPS 2025*.
- [2] T. Qiu*, **Z. He***, T. Chugh, M. Kleiman-Weiner (2025). The Lock-in Hypothesis: Stagnation by Algorithm. *ICML 2025*.
- [3] **Z. He***, T. Qiu*, T. Lin, M. Glickman, J. Wihbey, M. Kleiman-Weiner (2025). Position: AI Systematically Rewires the Flow of Ideas. *ICLR 2025 BiAlign Workshop*.
- [4] **Z. He***, M. Tehenan*, J. Achterberg, K. Collins, K. Nejad, D. Akarca, Y. Yang, W. Gurnee, I. Sucholutsky, Y. Tang, R. Ianov, G. Ogden, C. Li, K. Sandbrink, S. Casper, A. Ivanova, G. W. Lindsay (2024). Multilevel interpretability of artificial neural networks: leveraging framework and methods from neuroscience.
- [5] T. Qiu, A. H. Ismail, **Z. He**, S. Feng (2026) Self-Improvement as Coherence Optimization: A Theoretical Account. *arXiv preprint arXiv:2601.13566*.
- [6] J. Ji, T. Qiu, B. Chen, B. Zhang, H. Lou, K. Wang, Y. Duan, **Z. He**, J. Zhou, Z. Zhang, F. Zeng, K. Y. Ng, J. Dai, X. Pan, A. O’Gara, Y. Lei, H. Xu, B. Tse, J. Fu, S. McAleer, Y. Yang, Y. Wang, S. C. Zhu, Y. Guo, W. Gao (2023). AI Alignment: A Comprehensive Survey. Accepted by ACM Computing Surveys.
- [7] A. Chan, R. Salganik, A. Markelius, C. Pang, N. Rajkumar, D. Krasheninnikov, L. Langosco, **Z. He**, Y. Duan, M. Carroll, M. Lin, A. Mayhew, K. Collins, M. Molamohammadi, J. Burden, W. Zhao, S. Rismani, K. Voudouris, U. Bhatt, A. Weller, D. Krueger, T. Maharaj (2023). Harms from increasingly agentic algorithmic systems. *Accepted by ACM FAccT 2023*

PROFESSIONAL SERVICES

Invited Talks:

- Nov 2025 META FAIR
- Nov 2025 MIT
- Oct 2025 UK AI Security Institute
- Oct 2025 University of Chicago
- Sep 2025 Tsinghua University
- Jul 2025 University of Washington
- Feb 2025 Cambridge University

Mentoring:

- Jul 2025 – Present Supervised Program for Alignment Research

- Jul 2025 – Oct 2025 Algoverse AI Safety Fellowship
- June 2026 - Cambridge MARS

Reviewing

Nov 2025 - IASEAI 2026

2025 Onwards - Transactions on Machine Learning Research (TMLR)

2026 NeurIPS